```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
song = pd.read_csv("/content/songdata.csv", on_bad_lines='skip', engine='python')
```

```
song.head()
```

	artist	song	link	text
0	ABBA	Ahe's My Kind Of Girl	/a/abba/ahes+my+kind+of+girl_20598417.html	Look at her face, it's a wonderful face \nAnd...
1	ABBA	Andante, Andante	/a/abba/andante+andante_20002708.html	Take it easy with me, please \nTouch me gentl...
2	ABBA	As Good As	/a/abba/as+good+as+new_20003033.html	I'll never know why I had to go \nWhy I had

```
song = song.drop(columns=['artist','link'])
```

```
song.isnull().sum()
```

	0
song	0
text	0

dtype: int64

```
from sklearn.feature_extraction.text import CountVectorizer
cv = CountVectorizer(max_features=34000, stop_words='english')
```

```
song['text'] = song['text'].str.lower()
```

```
song
```

	song	text
0	Ahe's My Kind Of Girl	look at her face, it's a wonderful face \nand...
2	As Good As New	i'll never know why i had to go \nwhy i had t...
4	Bang A Boomerang	making somebody happy is a question of give an...
863	Heimlich Maneuver	i love you so much i need the heimlich maneuve...
864	History Is Made By Stupid People	scott became famous for freezing to death in a...
865	I Am Cow	i am cow hear me moo \ni weigh twice as much a...
866	Pulled My Train	i fell in love in \nit's the very best in my...
867	Jesus' Brother Bob	if you haven't heard of me \ni wouldn't be su...

868 rows × 2 columns

```
song.shape
```

```
(868, 2)
```

```
song['text'][0]
```

```
'look at her face, it's a wonderful face  \nand it means something special to me  \nl
ook at the way that she smiles when she sees me  \nhow lucky can one fellow be?  \n
\nshe's just my kind of girl, she makes me feel fine  \nwho could ever believe that s
he could be mine?  \nshe's just my kind of girl, without her i'm blue  \nand if she e
ver leaves me what could i do, what could i do?  \n  \nand when we go for a walk in t
he park  \nand she holds me and squeezes my hand  \nwe'll go on walking for hours and
talking  \nabout all the things that we plan  \n  \nshe's just my kind of girl, she m
akes me feel fine  \nwho could ever believe that she could be mine?  \nshe's just my
kind of girl, without her i'm blue  \nandi if she ever leaves me what could i do, what
could i do?\n\n'
```

```
from sklearn.feature_extraction.text import CountVectorizer
cv = CountVectorizer(max_features=5000, stop_words='english')
```

```
vectors = cv.fit_transform(song['text']).toarray()
```

```
vectors[0]
```

```
array([0, 0, 0, ..., 0, 0, 0])
```

```
cv.get_feature_names_out()
```

```
array(['000', '103', '105', ..., 'yum', 'zone', 'zoo'], dtype=object)
```

```
!pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from r
```

```
import nltk
```

```
from nltk.stem.porter import PorterStemmer
ps = PorterStemmer()
```

```
def stem(text):
  a=[]
  for i in text.split():
    a.append(ps.stem(i))

  return ' '.join(a)
```

```
stem('loved to hell')
```

```
'love to hell'
```

```
song['text'] = song['text'].apply(stem)
```

```
song['text'][0]
```

```
'look at her face, it' a wonder face and it mean someth special to me look at the way
that she smile when she see me how lucki can one fellow be? she' just my kind of gir
l, she make me feel fine who could ever believ that she could be mine? she' just my k
ind of girl, without her i'm blue and if she ever leav me what could i do, what could
i do? and when we go for a walk in the park and she hold me and squeez my hand we'll
go on walk for hour and talk about all the thing that we plan she' just my kind of gi
rl, she make me feel fine who could ever believ that she could be mine? she' just my
kind of girl, without her i'm blue and if she ever leav me what could i do, what coul
d i do?'
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
cs = cosine_similarity(vectors)
```

```
sorted(list(enumerate(cs[0],)),reverse=True,key=lambda x:x[1])[1:6]
```

```
[(34, np.float64(0.5114296826843978)),
 (92, np.float64(0.4530341419930316)),
 (411, np.float64(0.45198917872784833)),
 (274, np.float64(0.2748972773956365)),
 (231, np.float64(0.2708021147284102))]
```

```
song[song['song'] =='Bang'].index[0]
```

```
np.int64(3)
```

```
song.iloc[0].song
```

```
'Ahe's My Kind Of Girl'
```

```
def Recommend(songs):
  song_index = song[song['song'] == songs].index[0]
  distance = cs[song_index]
  top_song = sorted(list(enumerate(distance)),reverse=True,key=lambda x:x[1])[1:6]
```

```
    for i in top_song:
```

```
Recommend('Bang-A-Boomerang')
```

```
Bang
Dum Dum Diddle
If You Love Me
I Love America
Falling In Love With Love
```

```
import pickle
```

```
pickle.dump(song.to_dict(),open('song.pkl','wb'))
```

```
pickle.dump(cs,open('cs.pkl','wb'))
```